

Proof of Concept (PoC): Cricket-Themed Team Performance Web Application

1. Objective

The purpose of this PoC is to validate the feasibility of a gamified team performance tracking system using a cricket theme. The PoC will focus on essential functionalities such as user roles, tournament tracking, and performance analysis to ensure a smooth and engaging experience for users.

2. Key Features to Validate

2.1 User Authentication & Roles

- Secure login for players and coaches.
- Role-based access control:
 - **Players:** View personal performance stats and tournament details.
 - **Coaches:** Manage tournaments, update player data, and oversee team activities.

2.2 Team Dashboard

- Display team group photo, name, and statistics.
- Provide an overview of active tournaments and past performance.

2.3 Tournament & Performance Tracking

- **Tournaments Page:**
 - List of activities/projects (e.g., MPC, Namoi AI, Namoi Transcript, Namoi Revamp).
 - Performance scores assigned to each project.
- **Scoreboard Page:**
 - Individual contributions.
 - Activity scores and rankings.
 - Courses and skill development progress.

2.4 Coach Control Panel

- Add, edit, and remove tournaments.
- Modify team details (team logo, group photo, and statistics).

- Update player performance metrics and generate reports.

3. Technical Feasibility

3.1 Technology Stack

Component	Technology
Frontend	React.js + TailwindCSS
Backend	Node.js + Express.js
Database	MongoDB (NoSQL)
Authentication	Login
Hosting	Vercel (Frontend), Render (Backend)

3.2 Data Flow

1. **User Login** → JWT authentication verifies role (Player/Coach).
2. **Dashboard Display** → Fetches team data and tournaments from MongoDB.
3. **Performance Updates** → Coaches update scores and project details.
4. **Player View** → Players access their individual performance statistics.

4. PoC Success Metrics

- **Functional UI:** A clickable prototype demonstrating navigation and dashboard elements.
- **Working Authentication:** Secure login with role-based access.
- **Basic Tournament & Score Tracking:** Ability to add/view/update performance data.
- **User Experience:** Intuitive interface with smooth transitions.

6. Next Steps

- Develop a wireframe/mockup using Figma.
- Implement a minimum viable version with core functionalities.
- Conduct user testing with a small group to gather feedback.
- Iterate and refine based on test results.

7. Conclusion

This PoC establishes the feasibility of the Cricket-Themed Team Performance Web Application by ensuring a functional, scalable, and user-friendly experience. The validation of key features will pave the way for full-scale development and deployment.